Structure Sensitive and Semantic Alignment Synergistic Enhancement Network for Remote Sensing Change Detection

Wei Wang, Huilin Ren, Xin Wang, and Ji Li

***Abstract*—Remote sensing (RS) image change detection (CD) provides an extremely effective technical means for urban expansion analysis, land use assessment, and other tasks. However, RSCD often faces the problem of uneven edges caused by ignoring structural continuity in pixel-level prediction, and pseudo-change interference caused by low-frequency color changes induced by light and shadow, which seriously affects detection accuracy and robustness. For this reason, we propose a structure sensitive and semantic alignment synergistic enhancement network (S^2ENet) for RSCD. Centered on structure preservation and semantic enhancement as its core design philosophy, the framework aims to precisely identify authentic changes while effectively suppressing background artifacts. Specifically, a hierarchical adaptive frequency–spatial synergistic change focusing (FSCF) module is designed to analyze high-frequency subbands of shallow features in the frequency domain, thereby adaptively enhancing edge structures. A learnable denoising mechanism is introduced to suppress high-frequency noise to offset artifacts. The contrastive locking mechanism is fused in the spatial domain to further accurately define the real change area and improve the perception sensitivity and positioning accuracy of structural changes. In addition, a multilayer bidirectional progressive fusion (MBPF) module is proposed to establish a shallow-to-deep structure guidance and a deep-to-shallow semantic feedback path, facilitate bidirectional multiscale information flow, and effectively achieve cross-layer feature collaborative modeling and semantic consistent expression. Rigorous evaluations across four widely recognized CD datasets (SYSU-CD, CLCD, WHU Building, and LEVIR-CD datasets) demonstrate that S^2ENet outperforms nine state-of-the-art (SOTA) methods. Relative to the suboptimal network, S^2ENet achieves intersection over union (IoU) improvements of 1.43%, 2.02%, 0.32%, and 0.37% on respective datasets. The code is available at https://github.com/ahaha-16/S2ENet**

***Index Terms*—Change detection (CD), frequency–spatial synergistic, multilayer bidirectional interaction, semantic alignment, structure sensitive.**

I. Introduction

REMOTE sensing (RS) image change detection (CD) aims to automatically identify spatial position and category changes in land cover or terrain status that occur within the same geographical area by analyzing RS images captured at different times. The spatial structure evolution and the demand for high-frequency monitoring driven by the acceleration of urbanization are promoting the core application of RSCD technology in urban expansion identification, land use assessment, and disaster prevention and control [1], [2], [3], [4], and continue to innovate with the development of high-resolution images. Early conventional approaches to RSCD predominantly relied on simple algebraic operation-based techniques, image transformation-based strategies, and image classification-based frameworks. Among these, algebraic methods, such as image difference method [5], image ratio method [6], change vector analysis method [7], and other algorithms, determine the changed pixels by performing common algebraic operations on RS images in different time phases. Transformation-based methods, such as principal component analysis (PCA) [8] and kauth-thomas transformation (KT) [9], project features into specific spaces to highlight changing pixels. Classification-based approaches, including support vector machine (SVM) [10] and random forest (RF) [11], detect changes by comparing the classification maps of bi-temporal images. Although conventional CD methods offer operational simplicity, their reliance on handcrafted feature extraction proves inadequate for characterizing the complex information in high-resolution RS images, frequently leading to high missed detections, error propagation, and ultimately suboptimal performance.

In recent years, the rapid accumulation of massive volumes of RS imagery has driven the widespread application of deep learning (DL) methods within the field of RS image processing [12], [13]. Leveraging their capabilities for automated feature extraction and high-level semantic representation, these methods have become indispensable core technologies in CD [14], [15]. Benefiting from their multilayer stacked architectures and hierarchical feature learning ability, convolutional neural networks (CNNs) have established dominance in CD [16], [17]. Early CNN-based networks [18] processed CD by first fusing the bi-temporal images and then feeding the combined input into the model. However, this approach fails to fully exploit the detailed information inherent within each individual temporal image. To address this limitation, Siamese networks [19], [20] are introduced, utilizing dual branches to independently explore the geospatial information contained within the pre-temporal and post-temporal images. Nevertheless, the advent of Siamese networks presents sig-

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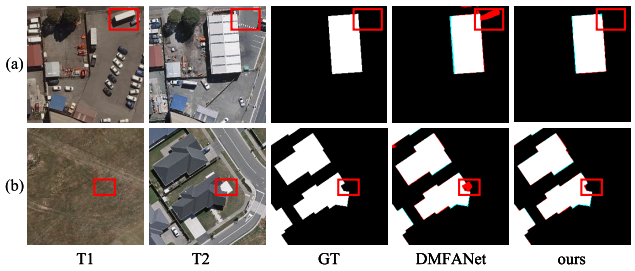


Fig. 1. Visual comparison with the latest method. T1 and T2 are bi-temporal images, and GT is the ground truth. White, black, red, and blue are used for TP, TN, FP, and FN, respectively. This clearly reflects the challenges in the field of CD. (a) Problem of uneven edges and pseudo-changes caused by temporary objects. (b) Problem of pseudo-changes caused by low-frequency color changes.

nificant challenges concerning the fusion of features extracted from these dual branches. Simple fusion strategies often prove inadequate to compensate for pseudo-changes induced by variations in illumination or seasonal conditions between bi-temporal images. Numerous studies [21], [22] have tackled the interaction of bi-temporal information within the spatial domain by leveraging attention mechanisms or stacking convolutional layers. For instance, DMINet [23] integrates self-attention with cross-attention to construct a novel attention mechanism specifically designed to suppress interference from irrelevant changes. SEIFNet [24] employs a dual-branch interaction module, stacking convolutional layers on the cascaded branch to enhance feature representation while simultaneously adopting an attention mechanism on the difference branch to focus on genuine changes. Changer [25] proposes exchange interactions conducted either space or channels to effectively mine true changes. Other research [26] explores the extraction of temporal information by utilizing the concept of long short-term memory (LSTM) networks [27]. Following the fusion of dual-branch features, the challenge of integrating multiscale features subsequently arises. While some approaches [18] resort to direct concatenation or element-wise addition of multiscale features for output generation, the majority of research [28], [29] adopt a strategy of unidirectional fusion involving progressive upsampling and concatenation layer by layer.

Although existing methods have made significant strides in suppressing pseudo-changes and enhancing feature discriminability, they continue to confront two persistent challenges, as illustrated in Fig. 1. On the one hand, the predominant focus on feature modeling and interaction within the spatial domain impedes the effective establishment of global structural constraints. This limitation renders the models susceptible to nonstructural interference from factors like illumination variations and seasonal shifts, which consequently enables low-frequency color changes to induce pseudo-changes. Concurrently, insufficient structural awareness hinders the model's ability to maintain spatial continuity in predictions, particularly within regions characterized by blurred or transitional edges, leading to unsmooth boundaries. On the other hand, prevalent multiscale feature fusion strategies predominantly rely on bottom-up unidirectional aggregation. This approach lacks bidirectional interaction between semantic and structural information, further compromising the expressive capacity for boundary structures. Consequently, pixel-wise predictions suffer from edge fragmentation and blurring, diminishing the model's sensitivity to subtle changes and localized objects.

Motivated by the aforementioned considerations, we propose a structure sensitive and semantic alignment synergistic enhancement network (S^2ENet) for RSCD. Unlike existing state-of-the-art (SOTA) methods such as SEIFNet [24] and DMFANet [30], which fail to recognize the importance of structural continuity in exploring the true edges of change, and FIMP [31], which enhances spatial details in the frequency domain but suffers from structural drift, our proposed S^2ENet uniquely achieves frequency–spatial cooperative interaction with explicit drift suppression and detail-guided semantically enhanced bidirectional multiscale fusion. Initially, bi-temporal images are fed into a backbone network to extract multiscale features. Subsequently, to enhance structural awareness for accurately locating genuine changes, a hierarchical adaptive frequency–spatial synergistic change focusing (FSCF) module is designed. This synergistic approach strategically applies frequency-domain analysis only to the first two layers, thereby fully leveraging the frequency domain's selective sensitivity to shallow structural information alongside the spatial domain's capacity for fine-grained positional constraints. Following this, a novel multiscale fusion framework, termed the multilayer bidirectional progressive fusion (MBPF) module, is devised. This module facilitates collaborative modeling of semantics and edges through bidirectional guidance. Finally, deep supervision is applied to the multiscale decoder outputs, and the shallow-level output is selected as the final result to produce the CD map. The principal contributions of this work are as follows.

1) To achieve multiperspective and multilevel mining of genuine changes for significantly enhanced structural continuity and semantic consistency, an S^2ENet is proposed, designed with the core principles of structure preservation and semantic augmentation.
2) To achieve highly sensitive capture of genuine differences and enhanced structural continuity in CD, the FSCF is designed. Within the frequency domain, this module significantly amplifies the response to authentic changes while smoothing edges by adaptively enhancing the high-frequency structural components of shallow features. Concurrently, a learnable denoising mechanism is introduced to suppress noise and artifacts in the frequency domain. Within the spatial domain, a contrastive mechanism is incorporated to precisely localize regions of true change, thereby mitigating error propagation potentially induced by frequency-domain reconstruction. This dual domain approach synergistically achieves enhanced structural saliency for changes.
3) To enhance cross-scale feature modeling capabilities and achieve deep synergy with semantically consistent representation across multilevel features, the MBPF is further proposed. This framework establishes dual complementary guidance pathways. Through progres-

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sively refined bidirectional interactions across multiple stages, this module facilitates the mutual flow of interlayer information. It effectively mitigates the challenges of fragmented semantics and isolated structural representations, synergistically enabling enhanced context modeling and greater robustness to scale changes.

The rest of this article is structured as follows. A brief review of related work is provided in Section II. The overall architecture and specific details of the proposed S^2ENet are introduced in Section III. Experimental analysis is then presented in Section IV. Finally, a concise conclusion is provided in Section V.

II. Related Work

A. Frequency-Domain Analysis in Image Processing

Frequency-domain analysis has been integrated into image processing [32], [33] due to the enhanced observability and processability of certain features within this domain, where frequency representations can more effectively delineate texture, shape, and structural characteristics. For instance, WavEnhancer [34] leveraged frequency-domain information for image enhancement, devising distinct optimization strategies for low-frequency and high-frequency components to augment feature expressiveness. FPIE [35] introduced a lightweight diffusion model to generate a compact frequency prior in the latent space, which is then leveraged by a wavelet-based transformer to guide the restoration process and suppress artifacts. FDNet [36] introduced a semantic classification model for RS operating exclusively in the frequency domain, utilizing the scaling properties of wavelet transform as an alternative to traditional downsampling to mitigate detail loss. FGNet [37] enhanced the SwinTransformer [38] architecture by incorporating frequency-domain information, enabling selective focus on critical frequency components for improved segmentation. MFGFNet [39] and FTransDF-Net [40] adopted relatively direct approaches by integrating frequency-domain processing into CD pipelines for difference enhancement. FASI-Net [41] proposed leveraging low-frequency information in shallow layers for semantic interaction while utilizing high-frequency components in deeper layers to guide difference enhancement, thereby assisting spatial CD. Crucially, while these methodologies concentrate on exploiting frequency-domain information to amplify specific frequency components for superior image processing outcomes, our goal is to introduce frequency-domain analysis into RSCD and explore a method that can enhance true changes between bi-temporal features.

An effective feature metric plays a fundamental role in distinguishing global and local changes, as demonstrated in multimetric learning studies [42]. To achieve more precise identification of genuine change boundaries, frequency–spatial interaction strategies have garnered significant research in CD. WS-Net++ [43] proposed a wavelet-based frequency–spatial interaction module. This method applied the wavelet transform to bi-temporal features separately, performed channel weighting to emphasize more critical subbands, and subsequently conducted subband-wise cross-interaction to reduce interference from irrelevant changes. While this approach considers leveraging frequency-domain information to supplement detailed features, it lacks a comprehensive exploration of subband information. FIMP [31] introduced a Fourier transform-based feature interaction module. It captured change-relevant frequency components while suppressing redundant ones within the frequency domain, employed difference attention to enhance true changes, and finally performed temporal change enhancement in the spatial domain. Although FIMP thoroughly explores frequency-domain information, the structural drift inherent in frequency transformation compromises the precise localization of change shapes and positions. GSSR-Net [44] addressed the positional interference caused by frequency transformation by utilizing spatial assistance and an offset field for accurate localization of true changes. It also extensively explored the frequency subbands derived from the wavelet transform. However, this method overlooked the substantial noise interference prevalent within high-frequency subbands. Addressing these collective challenges, the FSCF is proposed. This approach not only conducts a thorough exploration of subband information within the frequency domain while adaptively suppressing noise in high-frequency components but also ensures precise localization of genuine changes within the spatial domain. This hierarchical co-optimization between the two domains enables the network to enhance structural sensitivity and focus on real changes.

B. Exploration of Multiscale Fusion Frameworks

Traditional DL approaches for image processing [45], [46] relied exclusively on deep feature for prediction. While the deep layer encapsulates rich semantic information, progressive network depth inherently reduces spatial resolution, leading to the loss of precise positional details of ground objects. To reconcile rich contextual understanding with accurate localization, U-Net [47] enhanced the fully convolutional network (FCN) [48] by introducing a U-shaped architecture incorporating skip connections. This design progressively upsamples features during decoding while integrating encoder features from corresponding resolution levels via skip connections. Similarly, to achieve precise recognition and localization, Lin et al. [49] developed the feature pyramid network (FPN) for object detection. HR-Net [50] further advanced this concept by proposing a framework that sustains high-resolution representations throughout the network depth, concurrently fusing multiresolution features in later stages. This approach preserved spatial sensitivity while extracting semantically rich features. In recent years, the majority of CD methods [24], [29] have adopted U-shaped encoder–decoder architectures as their foundation. However, this prevalent bottom-up and layer-by-layer unidirectional fusion strategy during decoding inadequately addresses the complementary interdependencies across features from multiple hierarchical levels. Consequently, significant potential remains for further exploration into more effective multiscale representation learning.

To comprehensively explore contextual relationships across multiscale features, research on multiscale fusion frameworks has proliferated within the CD domain. SNUNet [19] constructed a densely connected

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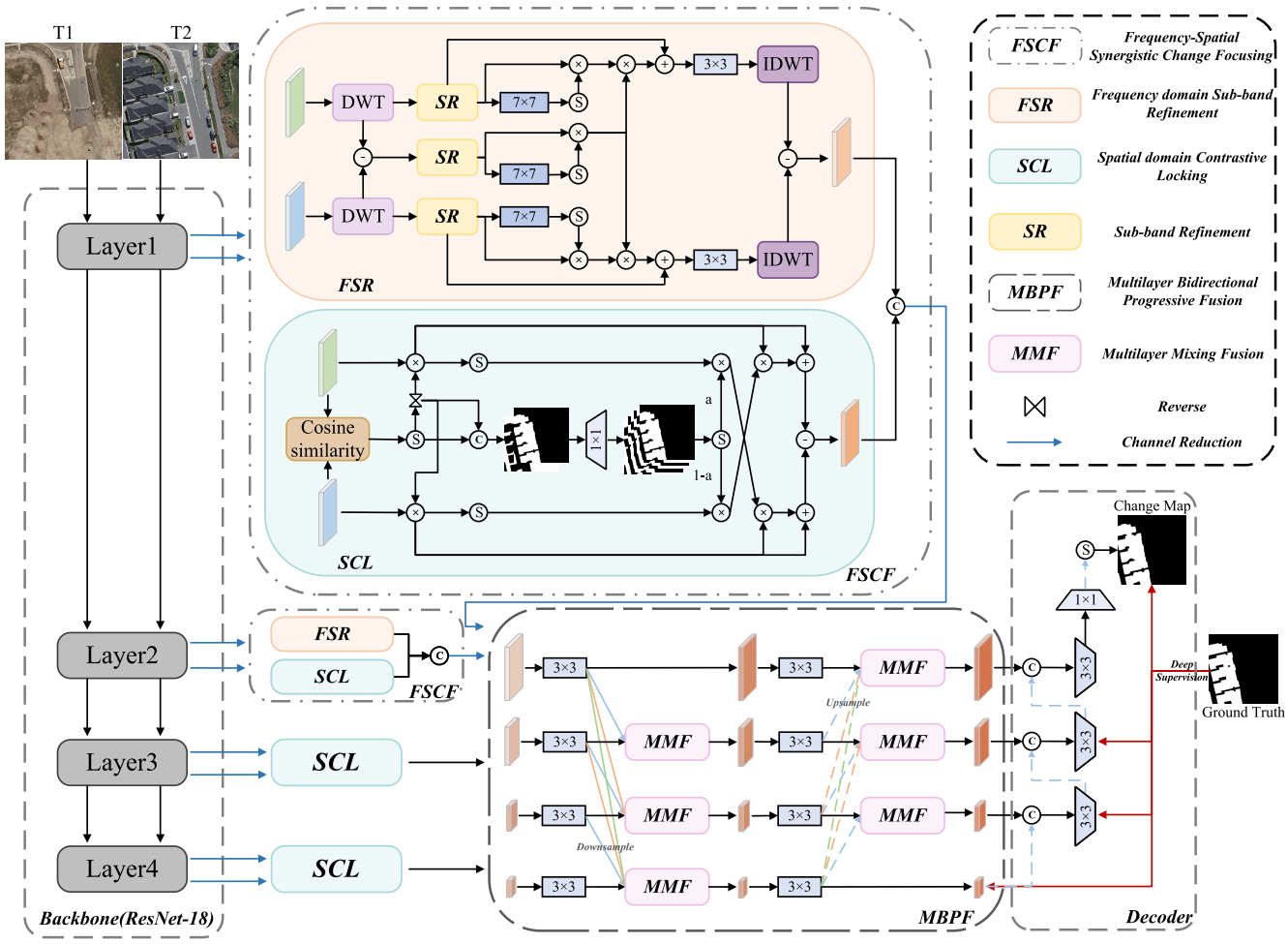


Fig. 2. Illustration of the S^{2}ENet we proposed, which is mainly composed of four principal components: a backbone network, the hierarchical adaptive FSCF module, the MBPF module, and a decoder.

multiscale feature fusion strategy to aggregate semantic information from different hierarchical levels. While this approach adequately explored contextual information across multiple layers, it fundamentally remained a unidirectional fusion decoding paradigm. Similarly, AFPF-Net [51] proposed progressively backward pairwise fusion of adjacent multilevel encoder features, culminating in a single merged feature. However, due to the absence of bidirectional interaction and structural guidance mechanism, this method often suffers from informational discontinuities in boundary regions, constraining its perception of fine-grained changes. FIMP [31] and DFFNet [52] incorporated a U-shaped structure within the decoder, achieving bidirectional layer-by-layer fusion of change features. This leveraged shallow-level detail to enrich deep semantic representations while simultaneously utilizing semantic information to strengthen discriminative capability at shallower levels. Although this established a bidirectional decoding framework, the complementary interplay of information across distinct hierarchical levels still holds potential for further enhancement. Addressing these limitations, the MBPF is proposed. This design fundamentally breaks from conventional unidirectional fusion paradigms. By implementing a top-down followed by bottom-up progressively refined bidirectional fusion strategy, it enables multistage interaction and iterative reprocessing of features across all hierarchical levels. This equips the CD network with significantly enhanced expressiveness and robustness.

III. Methodology

A. Architecture Overview

As illustrated in Fig. 2, the proposed network comprises four principal components: a backbone network, the hierarchical adaptive FSCF, the MBPF, and a decoder. The FSCF further integrates frequency-domain subband refinement (FSR) with spatial-domain contrastive locking (SCL). Bi-temporal images are first processed through a pretrained ResNet-18 [53] backbone to extract multiscale feature maps at four distinct resolutions: 128×128, 64×64, 32×32, and 16×16. Next, a 3×3 convolution with batch normalization (BN) and ReLU is applied to unify the channel dimension across all feature maps to 64. Afterward, to achieve highly sensitive capture of genuine differences and precise localization of global

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changes, bi-temporal features at the same scale are fed into the FSCF. This module tightly couples FSR with SCL, enabling bidirectional synergy between frequency-domain local edge focusing and spatial-domain global contrastive fusion. Given that shallow layers preserve richer spatial details [47] and that effective frequency-domain analysis requires sufficiently high-resolution feature maps, we apply the FSCF only to the first two shallow scales, while the deeper 32×32 and 16×16 layers use only the SCL submodule. The output features from the FSCF are again standardized to 64 channels using identical convolution operations. Recognizing that deep-level semantics are equally important as edges and textures, the MBPF is introduced. This module facilitates two rounds of alternating top-down and bottom-up iterative refinement, allowing information to be progressively polished throughout the network. This process significantly enhances sensitivity to subtle changes and ensures multiscale semantic consistency. Finally, the MBPF outputs undergo bottom-up fusion within the decoder to generate the final prediction. Deep supervision is applied to the multiscale feature maps during decoding to further optimize feature representation and produce an accurate CD result.

B. Hierarchical Adaptive FSCF Module

To achieve highly sensitive capture of structural continuity and precise localization of genuine differences in CD, a hierarchical adaptive frequency–spatial synergistic strategy is devised. Specifically, we construct the FSCF by employing local edge purification in the frequency domain and global contrastive fusion in the spatial domain and further propose the FSR and SCL modules. At the same time, considering that the deep features are not sensitive to details, only shallow features are analyzed in the frequency domain.

1) Frequency-Domain SR: Edges in complex RS images carry the most critical geometric and shape information. Compared to color or texture changes in homogeneous regions, structural changes demonstrate superior stability against non-structural interference such as illumination variations and color shifts. Edge enhancement effectively amplifies these structural differences, heightening model sensitivity and discriminative ability toward genuine changes. Within the frequency domain, low-frequency components correspond to flat areas and slowly changing gray levels, whereas high-frequency constituents concentrate around edges, textures, and fine details. Therefore, selective amplification of high-frequency subbands enables efficient edge-focused enhancement. However, high-frequency subbands encapsulate both authentic edges and disruptive elements, including noise and compression artifacts. An adaptive denoising unit effectively mitigates such interference while preserving real structural edges. This foundational understanding motivates our FSR module, as shown in Fig. 2, and we first perform 2-D Haar discrete wavelet transform (DWT) on bi-temporal features F_1 and F_2, which are decomposed into four different frequency subbands $\{F^1_{\mathrm{LL}}, F^1_{\mathrm{LH}}, F^1_{\mathrm{HL}}, F^1_{\mathrm{HH}}\}$ and $\{F^2_{\mathrm{LL}}, F^2_{\mathrm{LH}}, F^2_{\mathrm{HL}}, F^2_{\mathrm{HH}}\}$, respectively. Such a process can be described as

$$\left\{F^i_{\mathrm{LL}}, F^i_{\mathrm{LH}}, F^i_{\mathrm{HL}}, F^i_{\mathrm{HH}}\right\} = \mathrm{DWT}\left(F_i\right), \quad i \in \{1, 2\} \tag{1}$$

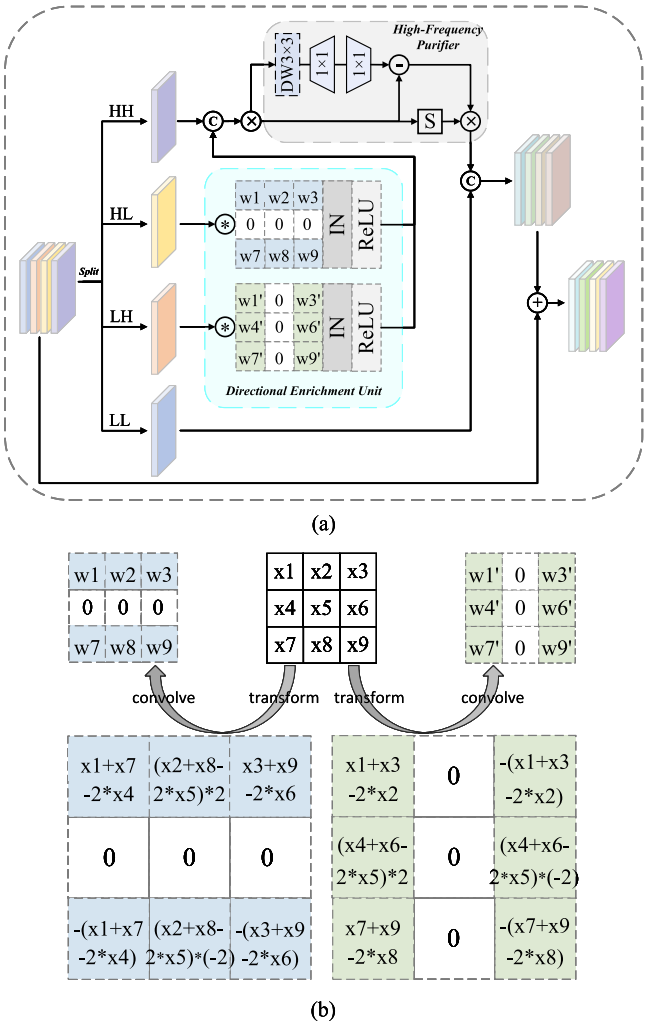


Fig. 3. Illustration of (a) SR module and (b) group of edge-aware difference convolutions.

where $F^i_{\mathrm{LL}}, F^i_{\mathrm{LH}}, F^i_{\mathrm{HL}}$, and F^i_{HH} correspond to low-frequency subband, vertical high-frequency subband, horizontal high-frequency subband, and diagonal high-frequency subband, respectively; and i is the time phase.

Next, each frequency subband is subtracted correspondingly to construct the difference branch to obtain the difference feature frequency map $\{F_{\mathrm{LL}}, F_{\mathrm{LH}}, F_{\mathrm{HL}}$, and $F_{\mathrm{HH}}\}$. Both the original and the difference frequency features undergo simultaneous subband refinement (SR), with the detailed workflow illustrated in Fig. 3(a). For the valuable high-frequency edge information and the noise artifacts in the frequency information, the directional enhancement unit and the high-frequency purifier are designed, respectively. Specifically, in the directional enhancement unit, we design a group of edge-aware difference convolutions with a kernel size of 3. These convolutions employ two orthogonal adjustable kernels to amplify the frequency response in the corresponding direction, focusing on horizontal and vertical edges. Fig. 3(b) shows this group of convolutions, which organically combines the concepts of the Sobel operator and pixel difference convolution [54]. It integrates the fixed gradient direction prior with the learnable

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difference mechanism, which not only retains the direction sensitivity of Sobel operator but also endows the network with the ability to adaptively optimize the edge response. During convolution operations, the original pixels in the local feature map patch covered by the convolution kernel are replaced by pixel differences. Such a process can be described as

$$X = \begin{bmatrix} x_1 & x_2 & x_3 \\ x_4 & x_5 & x_6 \\ x_7 & x_8 & x_9 \end{bmatrix} \tag{2}$$

$$r = \begin{bmatrix} 1 & -2 & 1 \end{bmatrix},\ u = \begin{bmatrix} 1 & 0 & -1 \end{bmatrix},\ D = \mathrm{diag}\,(1, 2, 1) \tag{3}$$

$$X_V = u^T\,(rXD),\ X_H = \left(DXr^T\right) u \tag{4}$$

where X denotes the original pixels in the local patch covered by the convolution kernel. We treat it as a matrix and apply a series of matrix transformations to obtain the pixel difference matrices X_V and X_H. These are then convolved with the kernel weights to generate the output features, enabling robust and accurate edge detection. diag$(\cdot)$ represents a diagonal matrix. In the high-frequency purifier, a lightweight denoising branch is designed to generate a learnable noise suppression feature for high-frequency coefficients. This branch effectively suppresses sensor noise and compresses artifacts while preserving the real structural details.

Recognizing significant redundancy between low-frequency components and spatial information, our refinement prioritizes high-frequency processing. SR employs horizontally sensitive convolution to enhance vertical edges in the HL subband, while vertically sensitive convolution targets horizontal edges in the LH subband. Directionally enhanced high-frequency components are then concatenated with the HH subband and passed through the denoising branch. This branch is implemented using 3×3 depth-wise (DW) separable convolutions and inverted bottleneck structures to produce a channel-wise noise suppression mask. Finally, the low-frequency component is concatenated with purified high-frequency output and integrated via residual connections with original frequency data. The following description details this procedure using difference feature refinement as the primary example:

$$D = \mathrm{Cat}\left(\mathrm{HEConv}\left(F_{\mathrm{LH}}\right), \mathrm{VEConv}\left(F_{\mathrm{HL}}\right), F_{\mathrm{HH}}\right) \tag{5}$$

$$H = \mathrm{Sign}\,(D) \times \left| D - \mathrm{Conv}_1^{(2)}\left(\mathrm{DWConv}_3\,(D)\right) \right| \tag{6}$$

$$R = F + \mathrm{Cat}\left(F_{\mathrm{LL}}, H\right) \tag{7}$$

where HEConv$(\cdot)$ and VEConv$(\cdot)$ represent horizontal edge-aware convolution and vertical edge-aware convolution with instance normalization (IN) and ReLU operations, respectively. Cat$(\cdot)$ denotes the channel-wise concatenation operation. $\mathrm{DWConv}_3(\cdot)$ represents a DW convolution with a kernel size of 3 and group normalization (GN). $\mathrm{Conv}_1^{(\mathrm{m})}(\cdot)$ represents a stack of m 1×1 convolutions with BN and ReLU, and Sign$(\cdot)$ is the sign function. $F = \{F_{\mathrm{LL}}, F_{\mathrm{LH}}, F_{\mathrm{HL}}, F_{\mathrm{HH}}\}$.

To holistically strengthen continuous edges or texture blocks, we introduce a 7×7 convolution after SR to generate an attention map, which is then multiplied with the original features. Subsequently, the difference branch is utilized to separately enhance temporal discrepancies for bi-temporal features and combine them with the corresponding SR feature through the residual connection. The bi-temporal features after SR are denoted as R_1 and R_2, respectively, with the overall process expressed as

$$\mathrm{diff}^f = R \times \left(\sigma\left(\mathrm{Conv}_7\,(R)\right)\right) \tag{8}$$

$$R_i' = R_i + \mathrm{diff}^f \times \left(R_i \times \left(\sigma\left(\mathrm{Conv}_7\left(R_i\right)\right)\right)\right),\quad i \in \{1, 2\} \tag{9}$$

where σ denotes the sigmoid operation and diff^f represents the difference-enhanced feature generated by the difference branch. We employ 3×3 convolutions with BN and ReLU activation to further refine the features from both temporal branches after the residual connection. Subsequently, the inverse DWT (IDWT) reconstructs these dual-branch features back into the spatial domain. Pixel-wise subtraction is then applied to produce the output of the FSR module. This process is formulated as

$$O^f = \left| \mathrm{IDWT}\left(\mathrm{Conv}_3\left(R_1'\right)\right) - \mathrm{IDWT}\left(\mathrm{Conv}_3\left(R_2'\right)\right) \right|. \tag{10}$$

2) Spatial-Domain Contrastive Locking: The downsampling inherent in DWT frequency analysis, coupled with nonlinear operations such as threshold denoising and directional enhancement, disrupts phase coherence among frequency coefficients. This degradation causes subtle spatial misalignment when reconstructing features back into the spatial domain via IDWT, potentially misinterpreting marginally displaced edges or contours as false changes. To address this limitation, a concurrently operating SCL module is introduced, which precisely calibrates feature response locations following frequency-domain enhancement. As depicted in Fig. 2, our approach begins by employing cosine similarity to construct a difference mask between bi-temporal features F_1 and F_2, which subsequently undergoes multiplicative fusion with the original features

$$\mathrm{sim} = \sigma\left(\delta\left(F_1, F_2\right)\right) \tag{11}$$

$$\mathrm{diff} = 1 - \mathrm{sim} \tag{12}$$

$$F_i' = F_i \times \mathrm{diff}^s,\quad i \in \{1, 2\} \tag{13}$$

where $\delta(\cdot)$ denotes the cosine similarity computation. Subsequently, to amplify genuine change boundaries for precise localization, the similarity map and the difference map are concatenated. A 1×1 convolution adjusts channel dimensionality to construct a foreground–background contrastive attention feature. Guided by this attention mechanism, we devise a dual-branch complementary enhancement framework with weighted modulation and cross-branch mutual guidance. This architecture dynamically regulates difference amplification and background suppression through dual-polarity weighting while introducing cross-branch adversarial perception via reciprocal modulation. Ultimately, residual connections robustly fuse the bi-temporal outputs, significantly boosting the model's sensitivity to authentic changes and its resilience against spurious differences. This operational sequence can be formally described as

$$\mathrm{diff}^s = \sigma\left(\mathrm{Conv}_1\left(\mathrm{Cat}\left(\mathrm{sim}, \mathrm{diff}\right)\right)\right) \tag{14}$$

$$O_1^s = F_1' + F_1' \times \left(\sigma\left(F_2'\right) \times \left(1 - \mathrm{diff}^s\right)\right) \tag{15}$$

$$O_2^s = F_2' + F_2' \times \left(\sigma\left(F_1'\right) \times \mathrm{diff}^s\right) \tag{16}$$

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where diff^s represents the foreground–background contrastive attention feature. Finally, a pixel-wise subtraction operation is performed on the contrast-enhanced bi-temporal features to obtain the output of the SCL module

$$O^s = \left| O_1^s - O_2^s \right|. \tag{17}$$

To balance detail preservation and computational efficiency, a hierarchical adaptive frequency–spatial synergistic strategy is employed. This strategy applies joint frequency–spatial fusion at shallow levels and utilizes exclusive spatial modeling in deeper layers

$$O_l = \begin{cases} \text{Cat}\left(O_l^f, O_l^s\right), & l \in \{1,2\} \\ O_l^s, & l \in \{3,4\} \end{cases} \tag{18}$$

where O_l^f denotes the output of the FSR module for the features of the lth layer and O_l^s represents the output of the SCL module. These outputs are concatenated along the channel dimension to constitute the final output of the FSCF.

C. MBPF Module

In multiscale feature fusion tasks, conventional unidirectional fusion often results in insufficient transmission of information flow and weak interlayer coupling. This problem becomes more pronounced when dealing with complex structures or small targets, leading to semantic discontinuities and detail loss. To overcome the constraints, the MBPF is proposed, which enhances the network's representational capacity and robustness for multiscale objects. Specifically, it first employs a top-down fusion pathway. This process leverages the spatially rich detail information from shallow layers to guide deeper features, thereby augmenting their discriminative ability and localization accuracy. Subsequently, a bottom-up fusion pathway inversely propagates semantically rich, context-aware information in deeper layers back to shallower features. This process mitigates the inherent lack of global perception in initial shallow features. The module not only enhances the semantic expressiveness but also reinforces the spatial discriminability of deeper representations.

As illustrated in Fig. 2, we first process all multiscale features through a 3×3 convolutional layer to dynamically homogenize contributions across scales. Subsequently, the first stage of multiscale fusion integrates progressively enriched spatial details to refine high-level semantic representations. In this process, we adopt a progressive fusion strategy, where the output of the lth layer depends on the fusion of the previous l layers. To implement this, we design a multilayer mixing fusion (MMF) module. This operation can be formally expressed as

$$X_l' = \text{Conv}_3\left(X_l\right), \quad l \in \{1,2,3,4\} \tag{19}$$

$$X_l'' = \begin{cases} X_1', & l \in \{1\} \\ \text{MMF}\left(\text{Down}\left(X_1'\right), \ldots, X_l'\right), & l \in \{2,3,4\}. \end{cases} \tag{20}$$

Given the rich spatial information inherent in the first-level feature, it is retained directly as the first-scale output. Down(·) denotes downsampling features from preceding levels to match the spatial resolution of level l. Subsequently,

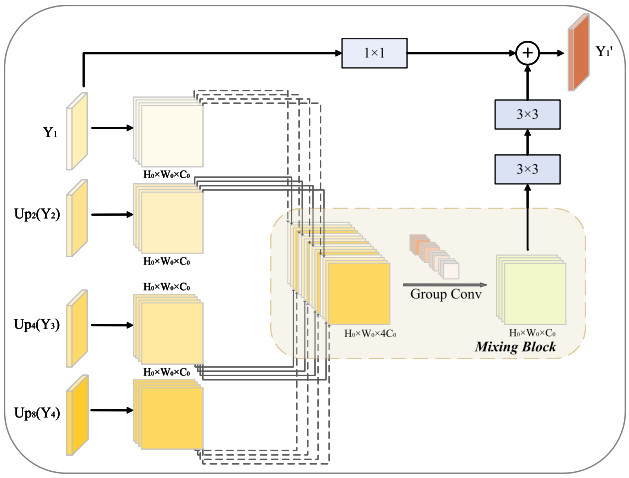


Fig. 4. Illustration of MMF module.

a 3×3 convolution dynamically modulates the representational weight of each fused feature scale. The second fusion stage similarly employs a progressive strategy but operates along an ascending pathway; the output at layer l incorporates fused information from the later l-layers

$$Y_l = \text{Conv}_3\left(X_l''\right), \quad l \in \{1,2,3,4\} \tag{21}$$

$$Y_l' = \begin{cases} Y_4, & l \in \{4\} \\ \text{MMF}\left(Y_l, \ldots, \text{Up}\left(Y_4\right)\right), & l \in \{1,2,3\}. \end{cases} \tag{22}$$

Mirroring the initial fusion stage, the deepest level feature, possessing strong semantic expressiveness, is preserved directly as the fourth-scale output. Up(·) denotes upsampling features from subsequent levels to match the spatial resolution of the lth layer. This bidirectional progressive strategy, first descending then ascending, enables multilayer features to undergo iterative cross-stage interaction and refinement.

Simplistic channel-wise concatenation, where cross-channel interaction occurs incidentally through subsequent convolutions, cannot ensure thorough fusion and alignment across scales. Therefore, we design the MMF to facilitate mutual perception and restructuring between feature pairs or groups. Through hierarchical alignment and weighted fusion, it maximizes mutual information across scales, enhancing the capture of cross-scale spatial and semantic dependencies for richer contextual modeling. Taking the shallow-layer MMF in the second fusion stage as an example, as depicted in Fig. 4, the current-layer feature $Y_1 \in \mathbb{R}^{H_0 \times W_0 \times C_0}$ is combined with upsampled features from the subsequent three levels. These inputs are first channel-mixed to produce a mix feature with $4C_0$ channels. Group convolution, with C_0 groups and a 3×3 kernel, is then applied to intensify mutual information. The output undergoes further processing via two 3×3 convolutions to maximize cross-scale cooperative representation. Finally, this enhanced representation is integrated with the original lth feature via a residual connection to yield the MMF output. This process can be formally expressed as

$$M = \text{GConv}_3\left(\text{Mix}\left(Y_1, \text{Up}_2\left(Y_2\right) \text{Up}_4\left(Y_3\right), \text{Up}_8\left(Y_4\right)\right)\right) \tag{23}$$

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$$Y_1' = \mathrm{Conv}_1(Y_1) + \mathrm{Conv}_3^{(2)}(M) \tag{24}$$

where $\mathrm{Up}_s(\cdot)$ denotes spatial upsampling by upsampling factor of s, $\mathrm{Mix}(\cdot)$ represents channel mixing operation, and $\mathrm{GConv}_3(\cdot)$ represents a 3×3 group convolution with C_0 groups.

D. Loss Function

CD inherently suffers from severe class imbalance, where the vast majority of pixels belong to the unchanged category, while changed regions are typically sparse and irregularly distributed. Training models directly with conventional pixel-level loss functions under this imbalance inevitably induces a bias toward the majority class, resulting in compromised detection capability for changed areas. To mitigate this limitation, we utilize the BCEDice loss as our composite loss function. This formulation organically combines binary cross-entropy (BCE) loss [55] and Dice loss [56], leveraging their complementary strengths to enhance robustness against class imbalance

$$\mathcal{L}_{\mathrm{bce}}(p, g) = g \times \log p + (1 - g) \times \log(1 - p) \tag{25}$$

$$\mathcal{L}_{\mathrm{dice}}(p, g) = 1 - \frac{2 \times g \times p}{\|g\| + \|p\|} \tag{26}$$

$$\mathcal{L}_{\mathrm{total}}(p, g) = \sum_{l=1}^{4} \lambda \times \mathcal{L}_{\mathrm{bce}}(p_l, g) + (1 - \lambda) \times \mathcal{L}_{\mathrm{dice}}(p_l, g) \tag{27}$$

where p and g represent the change prediction map and corresponding ground truth (GT), respectively. The $L1$ norm is denoted by $\|\cdot\|$. p_l represents the change prediction map of the lth layer. λ is a parameter that balances $\mathcal{L}_{\mathrm{bce}}$ and $\mathcal{L}_{\mathrm{dice}}$, and subsequent experiments will be conducted to identify the suitable value of λ.

IV. Experiments

A. Datasets

S^{2}ENet is evaluated on four publicly available CD datasets.

1) SYSU-CD [57]: Comprising 20 000 pairs of very high-resolution (VHR) satellite images acquired between 2007 and 2014, SYSU-CD supports multicategory CD, including building construction, vegetation change, and marine development. Original images were cropped into 256×256 pixel patches, with a 6:2:2 ratio partitioning (12 000/4000/4000 patches) for training, validation, and testing, respectively, enabling comprehensive analysis across diverse change scenarios.

2) CLCD [58]: This dataset comprises 600 pairs of cropland change samples collected by Gaofen-2 in Guangdong Province, China, in 2017 and 2019, with a spatial resolution of 0.5–2 m, where each pair consists of two 512×512 images and a corresponding binary cropland change label. We further cropped the image into 256×256 patches to facilitate model training and randomly allocating them into training (1440), validation (480), and test sets (480).

3) WHU-CD [59]: This urban building CD dataset features coregistered aerial images (0.075-m/pixel resolution) captured in 2012 and 2016, spanning $32\,507 \times 15\,354$ pixels with building construction or demolition changes. To facilitate model training, we subdivided the image into nonoverlapping 256×256 pixel patches, randomly allocating them into training (5947), validation (743), and test sets (744). This processing strategy preserves spatial details while enhancing training efficiency.

4) LEVIR-CD [60]: Designed for building CD, LEVIR-CD contains 637 pairs of 1024×1024 pixel images (0.5-m resolution) covering Texan urban areas between 2002 and 2018. To accommodate DL frameworks and reduce computational demands, we cropped original images into nonoverlapping 256×256 pixel subimages, yielding 7120 training pairs, 1024 validation pairs, and 2048 test pairs.

B. Implementation and Metrics

1) Implementation Details: S^{2}ENet was implemented in PyTorch and trained on NVIDIA A800 GPUs. Input data augmentation included random cropping, flipping, and temporal swapping of image pairs. We employed the Adam optimizer ($\beta 1 = 0.9$, $\beta 2 = 0.99$, and weight decay = 0.0001) with poly learning rate scheduling (power = 0.9 and maximum iterations = 40 000). Initial learning rate and batch size were set to 0.0005 and 32, respectively.

2) Evaluation Metrics: Performance was assessed using five standard metrics, including precision (Pre), recall (Rec), $F1$-score ($F1$), intersection over union (IoU), and Kappa coefficient (κ)

$$\mathrm{Pre} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}} \tag{28}$$

$$\mathrm{Rec} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}} \tag{29}$$

$$F1 = \frac{2 \times \mathrm{Pre} \times \mathrm{Rec}}{\mathrm{Pre} + \mathrm{Rec}} \tag{30}$$

$$\mathrm{IoU} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP} + \mathrm{FN}} \tag{31}$$

$$\mathrm{OA} = \frac{\mathrm{TP} + \mathrm{TN}}{\mathrm{TP} + \mathrm{FP} + \mathrm{TN} + \mathrm{FN}} \tag{32}$$

$$P_e = \frac{(\mathrm{TN} + \mathrm{FN}) \times (\mathrm{TN} + \mathrm{FP})}{(\mathrm{TP} + \mathrm{FP} + \mathrm{TN} + \mathrm{FN})^2} + \frac{(\mathrm{FP} + \mathrm{TP}) \times (\mathrm{FN} + \mathrm{TP})}{(\mathrm{TP} + \mathrm{FP} + \mathrm{TN} + \mathrm{FN})^2} \tag{33}$$

$$\kappa = \frac{\mathrm{OA} - P_e}{1 - P_e} \tag{34}$$

where TP, TN, FP, and FN represent the number of true positives, true negatives, false positives, and false negatives, respectively.

C. Comparison With SOTA Methods

S^{2}ENet is benchmarked against nine advanced CD approaches: FC-EF [18], SNUNet [19], ChangeFormer [61], DMINet [23], AFPF-Net [51], SEIFNet [24], FIMP [31], STRobustNet [62], and DMFANet [30]. As detailed in Table I, these methods represent diverse technical advancements with similar objectives to our framework. Crucially, all comparative models were retrained under identical experimental conditions according to the official model codes provided by the authors to ensure a fair comparison.

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TABLE I
DETAILED DESCRIPTION OF NINE COMPARISON METHODS

Methods	Year	Detailed Description
FC-EF [18]	2018	FC-EF employs an early fusion strategy, concatenating image pairs before processing through a fully convolutional network.
SNUNet [19]	2021	SNUNet introduces densely connected multi-scale fusion to enhance complementary interactions between detailed and semantic features. Neglecting the effects of pseudo-changes.
ChangeFormer [61]	2022	ChangeFormer implements a weight-shared siamese architecture with pure Transformer encoders and MLP decoders for superior global context modeling.
DMINet [23]	2023	DMINet develops a novel JointAtt mechanism to capture authentic bi-temporal changes, utilizing dual branches for feature fusion. An aggregation module for multi-scale difference information is constructed using incremental aggregation method.
AFPF-Net [51]	2024	AFPF-Net uses the difference feature of the previous scale to guide the aggregation of current scale features, and construct a multi-scale fusion strategy for progressive fusion of adjacent layers. Only spatial domain fusion.
SEIFNet [24]	2024	SEIFNet constructs a local global feature interaction module using convolution and attention mechanisms, and fully explored the contextual information between multi-scale features in the decoder. Only considering the extraction of spatial details through cascade convolution in the spatial domain.
FIMP [31]	2024	FIMP establishes feature interaction in both frequency and spatial domains, and a unique U-shaped multi-scale fusion strategy is proposed. Neglecting the structural drift inherent in frequency transformation.
STRobustNet [62]	2025	STRobustNet generates temporally and spatially consistent feature representations for identical land cover types across different regions and time periods.
DMFANet [30]	2025	DMFANet incorporates cross-layer context fusion and cross-temporal interaction during feature extraction, and a difference extraction module based on gating and global aggregation is constructed. Not aware of the importance of structural sensitivity for detecting real changes.

TABLE II
QUANTITATIVE COMPARISONS ON CD RESULTS. THE BEST AND SECOND-BEST RESULTS ARE HIGHLIGHTED IN RED AND BLUE, RESPECTIVELY

Methods		FC-EF	SNUNet	ChangeFormer	DMINet	AFPF-Net	SEIFNet	FIMP	STRobustNet	DMFANet	ours
Params.(M)		1.35	12.04	29.75	6.24	14.96	27.91	2.43	10.55	6.88	17.59
FLOPs(G)		3.57	54.83	21.14	14.48	12.26	8.37	2.99	20.97	11.94	24.92
SYSU (%)	F1	76.73	80.59	79.99	82.69	82.17	82.79	82.78	79.62	82.08	83.77
	Pre	76.79	82.00	80.58	84.26	83.89	84.69	84.60	82.05	86.36	86.07
	Rec	76.68	79.23	79.41	81.18	80.52	80.98	81.05	77.32	78.19	81.58
	IoU	62.25	67.49	66.65	70.48	69.74	70.64	70.62	66.13	69.60	72.07
	Kappa	69.56	74.74	73.87	77.47	76.82	77.64	77.62	73.57	76.90	78.93
CLCD (%)	F1	61.41	68.51	69.86	75.07	74.70	74.45	73.05	69.90	73.19	76.63
	Pre	65.53	72.97	71.94	75.75	76.16	72.72	70.78	71.32	74.02	79.95
	Rec	57.78	64.55	67.90	74.40	73.36	76.27	75.48	68.54	72.38	73.57
	IoU	44.31	52.10	53.68	60.09	59.62	59.30	57.54	53.73	57.72	62.11
	Kappa	58.52	66.14	67.51	73.08	72.71	72.35	70.81	67.54	71.06	74.83
WHU (%)	F1	80.08	93.03	89.84	93.84	93.47	93.65	93.84	92.95	94.19	94.36
	Pre	88.36	95.44	91.55	93.62	94.11	95.13	95.11	94.47	95.59	95.67
	Rec	73.22	90.73	88.18	94.06	92.85	92.22	92.62	91.48	92.82	93.09
	IoU	66.78	86.96	81.55	88.40	87.75	88.07	88.40	86.83	89.01	89.33
	Kappa	79.34	92.74	89.42	93.59	93.21	93.40	93.59	92.67	93.95	94.13
LEVIR (%)	F1	78.57	91.11	90.22	91.04	91.13	91.01	91.29	90.90	91.45	91.67
	Pre	79.55	92.85	91.31	91.08	91.76	92.37	92.20	91.37	91.44	92.03
	Rec	77.62	89.43	89.15	91.01	90.50	89.69	90.39	90.44	91.45	91.30
	IoU	64.71	83.67	82.18	83.56	83.70	83.51	83.97	83.32	84.24	84.61
	Kappa	77.44	90.64	89.70	90.56	90.65	90.54	90.83	90.42	90.99	91.22

1) Quantitative Evaluation: Comprehensive quantitative assessments of all methods on the SYSU-CD, CLCD, WHU-CD, and LEVIR-CD datasets are presented in Table II. Results across all four datasets substantiate the competitive superiority of the proposed S^{2}ENet. On the challenging SYSU-CD dataset, S^{2}ENet achieves the highest scores in

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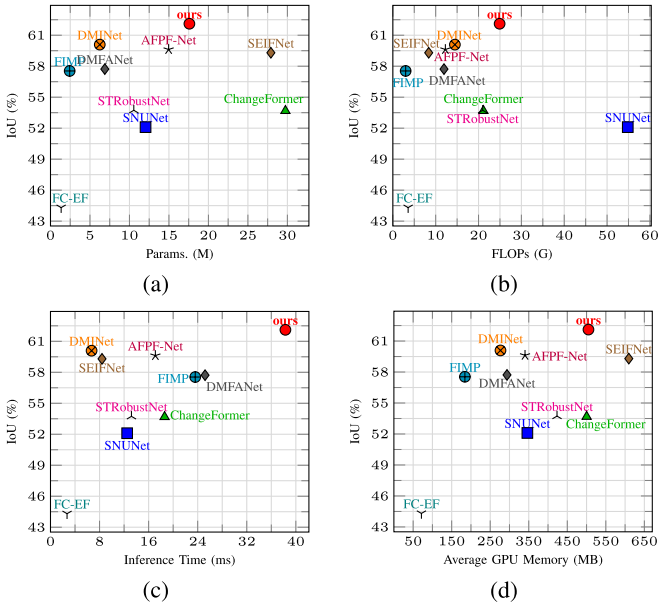


Fig. 5. Model complexity and running efficiency comparisons of different methods on the CLCD dataset. (a) Params. (b) FLOPs. (c) Inference time. (d) GPU memory.

$F1$ (83.77%), recall (81.58%), IoU (72.07%), and Kappa (78.93%). While DMFANet attains superior Precision on this dataset, indicating potential refinements for S^{2}ENet in exploring different information, our method demonstrates exceptional robustness. On the CLCD dataset, compared with the suboptimal network, S^{2}ENet increased $F1$ and IoU by 1.56% and 2.02%, respectively. The larger improvement mainly comes from the relatively simple structures of cropland scenes, where our structure sensitive and semantic alignment strategies can be more effectively leveraged. This further shows that our method is suitable for different types of datasets and possesses strong generalization capability. S^{2}ENet exhibits significant advantages in comprehensive metrics on the WHU-CD dataset. Although DMINet achieved the highest Recall here, indicating potential refinements for our approach in multilevel information fusion, our method still maintained a favorable balance between false positives and false negatives. On LEVIR-CD, S^{2}ENet further increased IoU and $F1$ to 84.61% and 91.67%, respectively, compared to the latest STRobustNet and DMFANet. This improvement can be attributed to S^{2}ENet's enhanced semantic modeling and structural sensitivity, enabling superior recognition of features overlooked by other methods and further validating the efficacy of frequency-domain analysis and bidirectional multiscale feature exploration. SNUNet achieved the highest precision on this dataset, which is potentially linked to the streamlined feature extraction of S^{2}ENet, and its high Pre also fully confirms the value of thorough multiscale fusion during decoding.

To holistically assess complexity, we select model parameters (Params) and floating-point operations (FLOPs), measuring parameter scale and computational workload. As indicated in Table II, the parameter and computational complexity of S^{2}ENet remain within an acceptable range. The introduced frequency analysis and bidirectional multiscale fusion enhance precision at a corresponding computational cost. However, compared with the latest DMFANet on the SYSU-CD dataset, our method improves $F1$ and Rec by 1.7% and 3.39%, respectively. Compared to the lightweight FIMP, S^{2}ENet demonstrates superior detection accuracy across all datasets despite higher complexity. In Fig. 5, we compare different methods on the CLCD dataset in terms of IoU, Params, FLOPs, inference time, and GPU memory. Unlike other approaches that either sacrifice accuracy for lower complexity or obtain higher accuracy at the cost of substantially increased computational demands, our method achieves the highest accuracy while keeping Params and FLOPs within a reasonable range. Furthermore, it attains a notable accuracy gain with only moderate increases in inference time and memory consumption. These quantitative results collectively attest to S^{2}ENet's effective balance of performance and efficiency, validating its overall effectiveness and superiority.

2) Qualitative Evaluation: Visual comparisons of different methods across four datasets are presented in Figs. 6–9, with prediction differences against GT highlighted using four distinct colors: white (TP), black (TN), red (FP), and blue (FN). Analysis of challenging samples reveals compelling advantages of our approach in three critical aspects.

a) Advantages in structural sensitivity: Many CD networks often neglect the structural integrity and edge smoothness of ground information due to pixel-level detection, resulting in internal voids and edge blurring problems. As shown in Figs. 6(a) and (b)–9(a) and (b), it is difficult for other methods to maintain the structural integrity of the true changes, resulting in certain false positives or omissions at the edges. In contrast, S^{2}ENet is sensitive to structure and maintains smooth edges. This indicates that our proposed FSCF effectively enhances the true edges in the frequency domain, while MBPF achieves complementary spatial and semantic information.

b) Advantages in detecting small target changes: In RSCD, the identification of small change targets has always been a challenge, primarily due to their extremely low pixel proportion, which makes them prone to losing spatial details during downsampling and vulnerable to interference from noise or background clutter. As demonstrated in Figs. 6(c) and (d)–9(c) and (d), compared to other methods, S^{2}ENet is more effective in detecting and accurately identifying small target changes. This result fully illustrates the advantages of spatial–frequency synergy and multiscale bidirectional complementary fusion in preserving spatial details, further validating the critical importance of detailed information in small target detection.

c) Advantages in suppressing pseudo-changes: A persistent challenge in RSCD is often caused by illumination changes and seasonal fluctuations that trigger irrelevant detections. As shown in Figs. 6(f), 8(e), and 9(e), highlights, transient objects, and shadows induce significant false alarms in most approaches, whereas S^{2}ENet substantially mitigates this issue. Numerous visual comparisons [Figs. 6(e) and (f)–9(e) and (f)] confirm our method's proficiency in discriminating authentic changes. This observation underscores the importance of structural preservation for highlighting genuine

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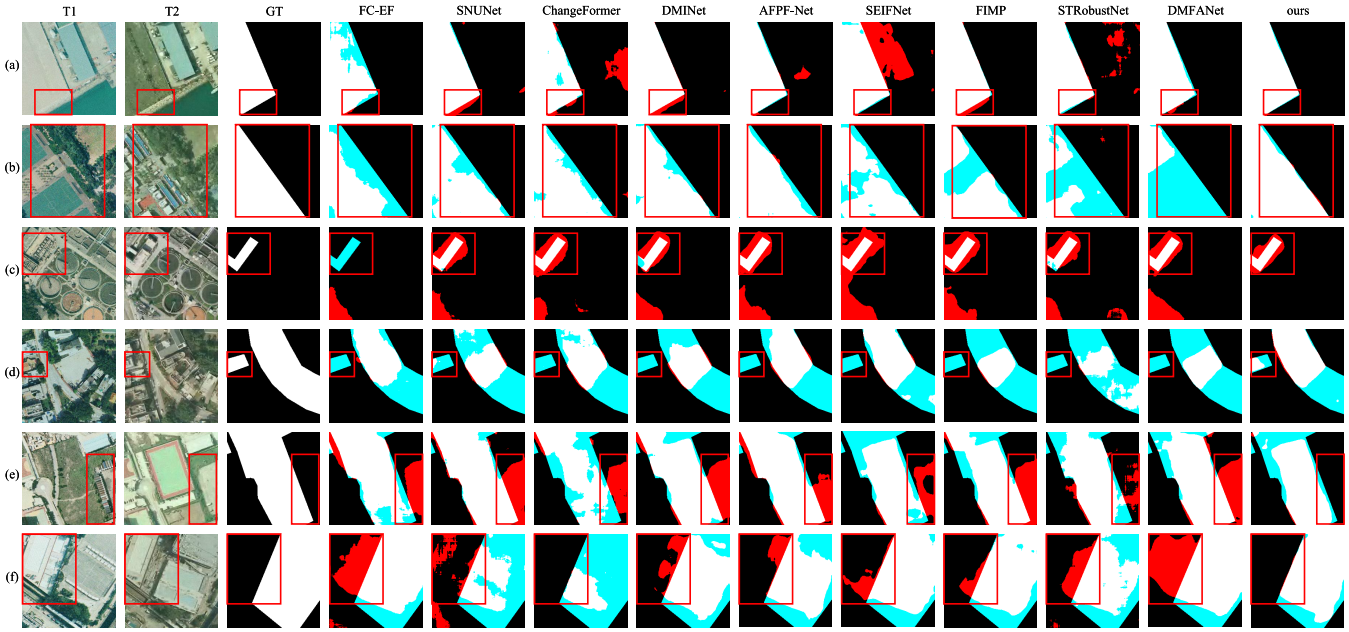


Fig. 6. Visualization results on SYSU-CD dataset. (a)–(f) Six representative samples. T1 and T2 are bi-temporal images, and GT is the ground truth. White, black, red, and blue are used for TP, TN, FP, and FN, respectively.

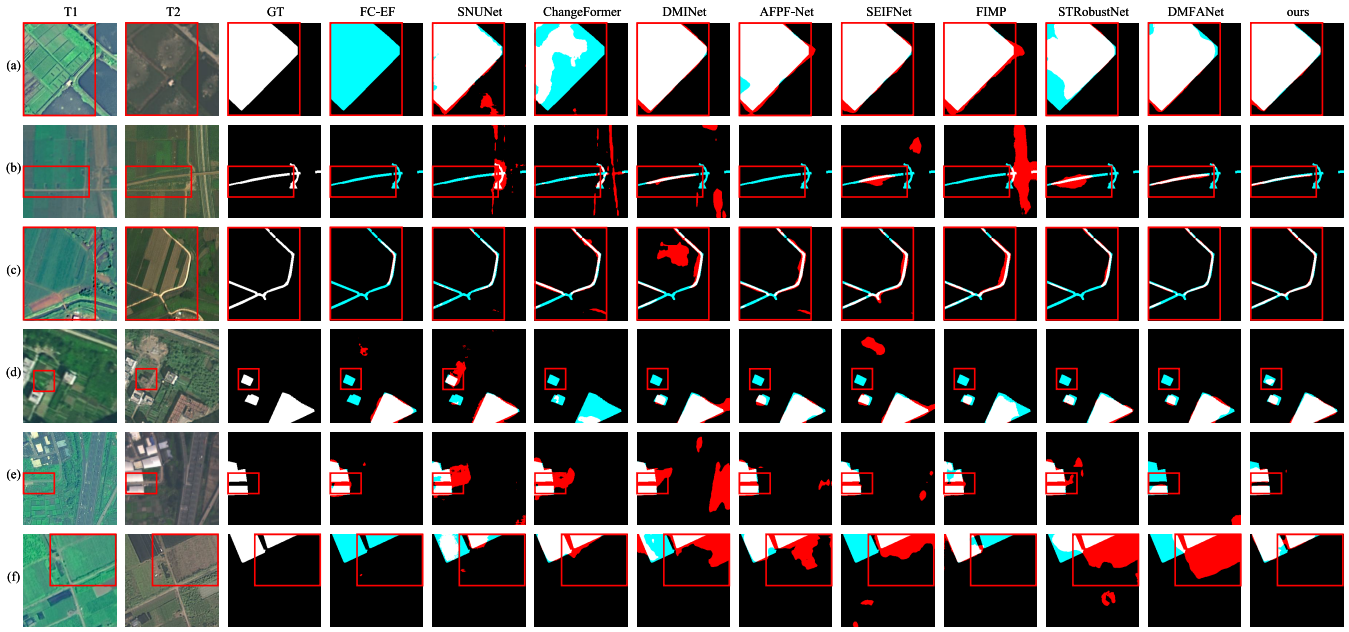


Fig. 7. Visualization results on CLCD dataset. (a)–(f) Six representative samples. T1 and T2 are bi-temporal images, and GT is the ground truth. White, black, red, and blue are used for TP, TN, FP, and FN, respectively.

changes, alongside enhanced semantic representation in shallow features and reinforced spatial discriminability in deep features, collectively improving relevant CD while suppressing noise.

Consequently, S^2ENet delivers compelling visualization results that align with the quantitative metrics in Table II, demonstrating consistently high-performance CD capability.

3) Cross-Dataset Generalization Evaluation: To evaluate cross-dataset generalization, we compare our approach with the two latest methods in Table III. The models were trained on LEVIR-CD, SYSU-CD, and CLCD, and tested on WHU-CD. Although all methods experience performance degradation due to domain shifts caused by factors such as sensor differences, resolution changes, and lighting conditions, our approach exhibits the smallest decline and achieves the best

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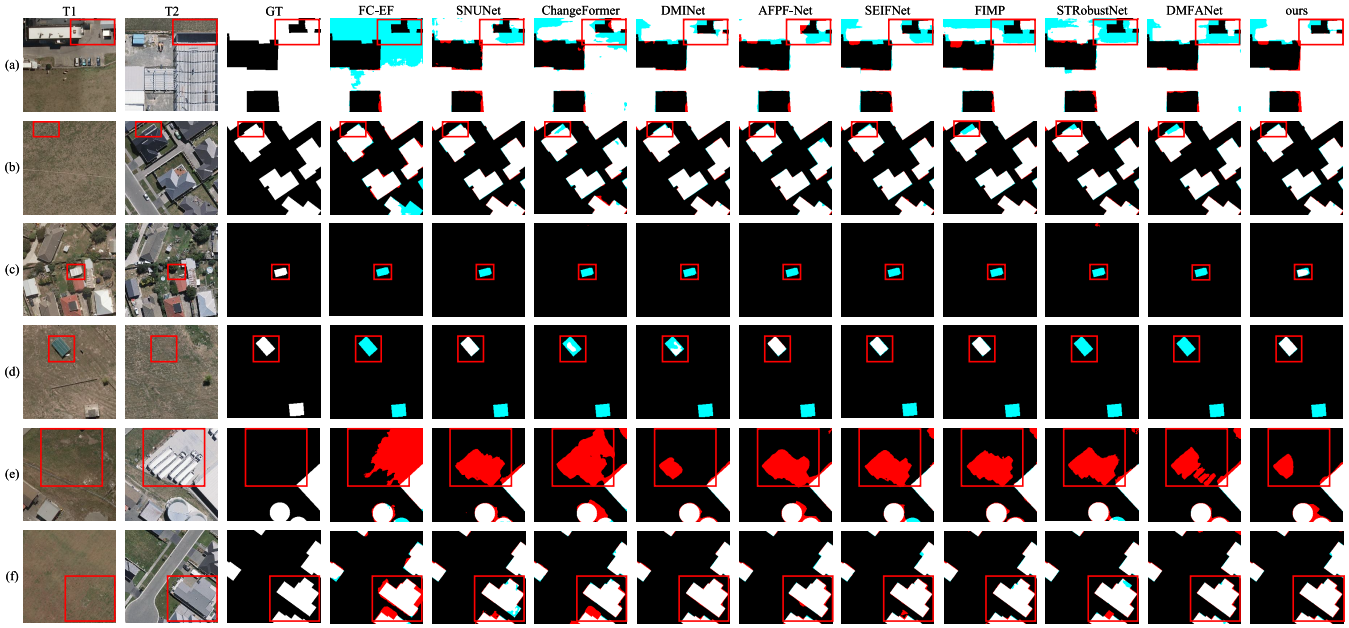


Fig. 8. Visualization results on WHU-CD dataset. (a)–(f) Six representative samples. T1 and T2 are bi-temporal images, and GT is the ground truth. White, black, red, and blue are used for TP, TN, FP, and FN, respectively.

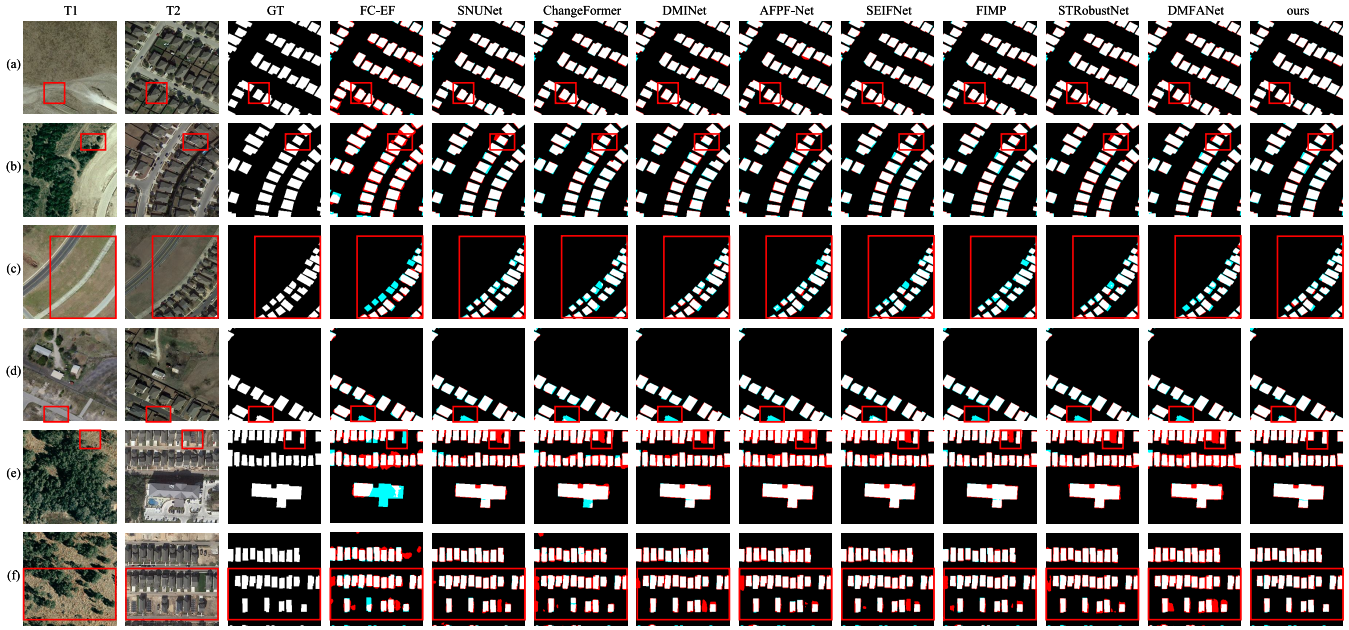


Fig. 9. Visualization results on LEVIR-CD dataset. (a)–(f) Six representative samples. T1 and T2 are bi-temporal images, and GT is the ground truth. White, black, red, and blue are used for TP, TN, FP, and FN, respectively.

results. Notably, it attains a 1.72% IoU improvement over the second-best performer, demonstrating superior robustness and generalization capability.

D. Ablation Study

To validate the effectiveness of the proposed S^2ENet components and configurations, we conduct comprehensive ablation experiments.

1) Effectiveness of FSCF: This module aims to construct a hierarchical adaptive frequency–spatial synergistic strategy, which performs frequency-domain analysis at shallow layers to enhance structural sensitivity and spatial semantic enhancement across all layers to focus on genuine changes. Ablating the entire FSCF (denoted as w/o FSCF in Table IV, #03) and simply concatenating bi-temporal features leads to significantly inferior performance compared to the S^2ENet (#01 in Table IV), confirming FSCF's overall contribution.

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TABLE III

QUANTITATIVE COMPARISONS OF CROSS-DATASET GENERALIZATION. MODELS ARE TRAINED ON LEVIR-CD, SYSU-CD, AND CLCD DATASETS AND TESTED ON THE WHU-CD DATASET. W/O MEANS WITHOUT. THE BEST RESULTS ARE HIGHLIGHTED IN RED

Methods	WHU (%)					
	w/o cross-dataset			cross-dataset		
	IoU	F1	Kappa	IoU	F1	Kappa
STRobustNet	86.83	92.95	92.67	54.68	70.70	69.43
DMFANet	89.01	94.19	93.95	60.53	75.42	74.39
ours	89.33	94.36	94.13	62.25	76.73	75.76

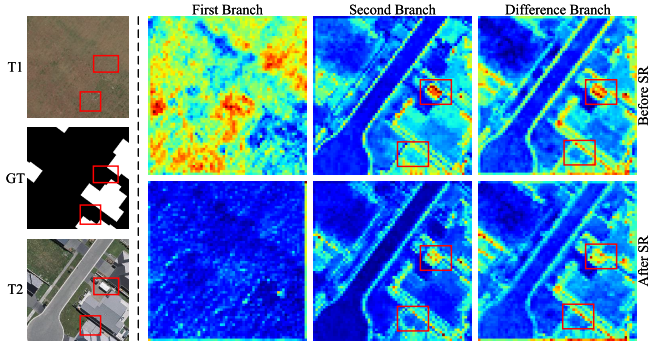


Fig. 10. Visualization results of feature maps before and after SR in the three branches of FSR. T1 and T2 are bi-temporal images, and GT is the ground truth. The first branch refers to the first branch in FSR. The second branch refers to the second branch in FSR. Difference branch refers to the difference branch in FSR. Before SR refers to the input feature map of SR. After SR refers to the output feature map processed by SR.

Further removing only the FSR component (#04 in Table IV) causes a notable performance drop, validating that frequency-domain analysis effectively enhances structural sensitivity for capturing authentic changes. To assess the adaptive hierarchy strategy, variants applying frequency–spatial synergistic only at the first layer (#05 in Table IV), the first three layers (#06 in Table IV), and all layers (#07 in Table IV) are tested. Quantitative comparisons demonstrate that restricting frequency–spatial synergistic primarily to the first two layers (#01) optimally balances effectiveness and computational efficiency. Ablating the SR component within FSR (#08 in Table IV) highlights the critical role of directional enhancement and high-frequency noise suppression in the frequency domain. To further verify the effectiveness of the proposed SR, we visualize the feature maps before and after SR in the three branches of FSR. As demonstrated in Fig. 10, the SR effectively suppresses noise artifacts while enhancing genuine change edges. Finally, removing the SCL module (#09 in Table IV) substantiates the importance of spatial-domain frequency shift correction for semantic enhancement.

2) Effectiveness of MBPF: This module aims to achieve cross-layer feature interaction and semantically consistent representation. Directly removing MBPF (#10 in Table IV) results in substantial performance degradation on SYSU-CD; IoU, $F1$, and Kappa decreased by 3.79%, 2.62%, and 3.27%, respectively, unequivocally demonstrating the efficacy of bidirectional progressive fusion. Replacing the MMF submodule

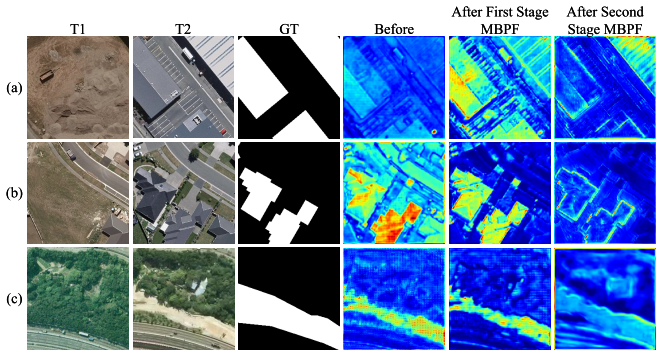


Fig. 11. Visualization results of feature maps before, during, and after bidirectional fusion in MBPF. (a)–(c) Three representative samples. T1 and T2 are bi-temporal images, and GT is the ground truth. Before refers to the input feature map of MBPF. After first-stage MBPF refers to the output feature map of the first round of multiscale progressive fusion from top to bottom. After second-stage MBPF refers to the output feature map of multiscale bidirectional progressive fusion.

within MBPF with simple concatenation (#11 in Table IV) also yields inferior results, emphasizing the necessity of thorough multilevel feature fusion and alignment. Furthermore, unidirectional fusion variants (#12 and #13 in Table IV) underperform the full bidirectional MBPF, confirming the advantage of bidirectional information flow. Similarly, we also visualize the input feature, the feature progressively fused from top to bottom, and the feature after bidirectional progressive fusion in the MBPF, as shown in Fig. 11. The MBPF enables the model to focus on real changes and enhances the structural saliency of these changes.

3) Discussion on the Synergistic Gain of FSCF and MBPF: As shown in Table IV, both FSCF (#10) and MBPF (#03) individually improved detection performance over the baseline (#02). However, their combination (#01) resulted in substantially greater gains, indicating that their functions are complementary rather than independent. Specifically, FSCF provides more discriminative input features for MBPF, while MBPF in turn better addresses the multiscale alignment issue.

4) Discussion on the Weight Parameter λ of Loss Function: To enhance the robustness of the network against class imbalance, we introduced the weight parameter λ to balance the contributions of BCE loss and Dice loss. The value of λ was constrained within the range [0, 1] with an interval of 0.25. As shown in Table V, when λ was set to 0.5, our model achieved the best performance across all four datasets, demonstrating that this setting provides the optimal balance between regional consistency and pixel-level calibration for strong generalization. Consequently, we adopt $\lambda = 0.5$ as the default configuration of our final model.

5) Discussion on Different Backbone Networks: We use ResNet-34 and ResNet-50 to further verify the proposed method. The results in Table VI show that with the deepening of the backbone network, its computational complexity has increased significantly, but its performance on SYSU-CD has decreased. This is because with the deepening of backbone networks, they are better at extracting higher level semantic features at the expense of spatial details, which further proves that CD is a structure sensitive task.

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TABLE IV
QUANTITATIVE COMPARISONS OF THE PROPOSED METHOD WITH DIVERSE SETTINGS ON FOUR DATASETS

No.	Variants	SYSU (%)			CLCD (%)			WHU (%)			LEVIR (%)		
		IoU	F1	Kappa	IoU	F1	Kappa	IoU	F1	Kappa	IoU	F1	Kappa
#01	ours	72.07	83.77	78.93	62.11	76.63	74.83	89.33	94.36	94.13	84.61	91.67	91.22
#02	w/o FSCF + MBPF	66.57	79.93	74.03	57.52	73.04	70.94	86.97	93.03	92.75	83.12	90.78	90.29
(a) Frequency-Spatial Synergistic Change Focusing Module (FSCF)													
#03	w/o FSCF	70.42	82.64	77.27	59.14	74.32	72.42	88.21	93.74	93.48	84.23	91.44	90.98
#04	w/o FSR	69.22	81.81	76.57	59.92	74.94	73.01	88.40	93.84	93.59	84.31	91.49	91.04
#05	FSR-1	71.41	83.32	78.34	61.45	76.12	74.20	88.12	93.69	93.43	84.01	91.31	90.85
#06	FSR-3	71.19	83.17	78.07	60.02	75.01	73.05	88.90	94.13	93.89	84.28	91.47	91.02
#07	FSR-4	71.45	83.35	78.28	58.52	73.84	71.75	89.00	94.18	93.94	84.05	91.33	90.87
#08	FSR-w/o SR	70.95	83.01	78.07	59.92	74.94	73.08	88.61	93.96	93.71	84.05	91.33	90.87
#09	w/o SCL	69.49	82.00	76.66	61.97	76.52	74.74	89.04	94.20	93.97	83.86	91.22	90.75
(b) Multilayer Bidirectional Progressive Fusion Module (MBPF)													
#10	w/o MBPF	68.28	81.15	75.66	59.71	74.78	72.83	88.85	94.10	93.86	83.58	91.05	90.58
#11	MBPF-w/o MMF	70.01	82.36	77.23	61.35	76.04	74.29	88.91	94.13	93.89	84.19	91.41	90.96
#12	MBPF-w/o up	71.39	83.31	78.29	60.92	75.71	73.87	88.74	94.03	93.79	84.05	91.34	90.88
#13	MBPF-w/o down	70.88	82.96	77.61	60.22	75.17	73.19	88.91	94.13	93.89	84.13	91.38	90.92

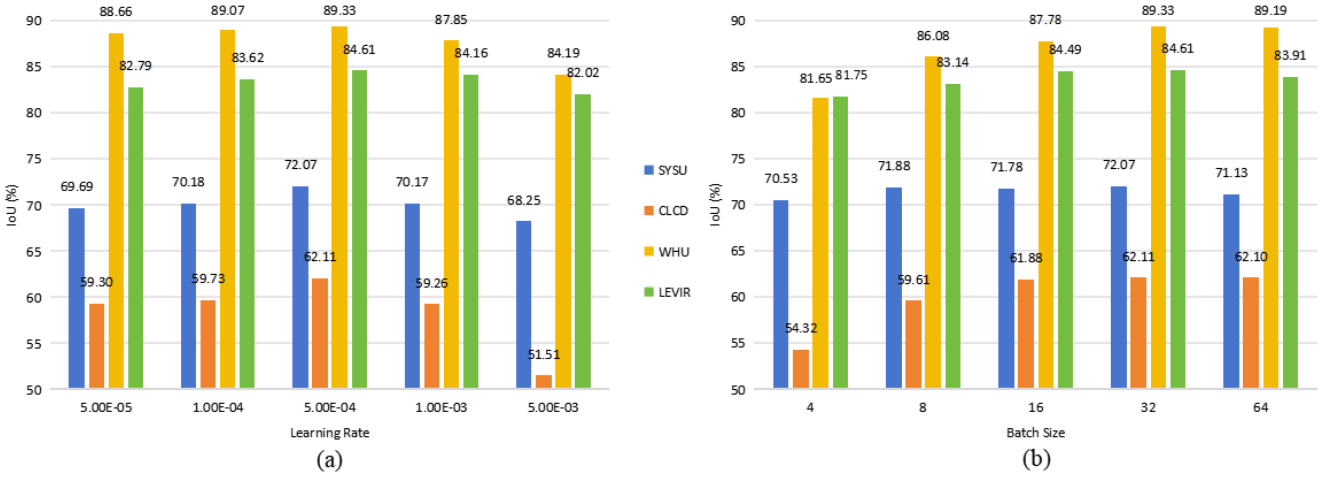


Fig. 12. Impact of different hyperparameters on IoU. (a) Learning rate. (b) Batch size.

TABLE V
QUANTITATIVE COMPARISONS OF DIFFERENT WEIGHT PARAMETERS λ OF LOSS FUNCTION ON FOUR DATASETS. THE BEST RESULTS ARE HIGHLIGHTED IN RED

λ	SYSU (%)		CLCD (%)		WHU (%)		LEVIR (%)	
	IoU	F1	IoU	F1	IoU	F1	IoU	F1
0.00	71.38	83.30	61.60	76.23	88.84	94.09	84.27	91.46
0.25	71.54	83.41	61.72	76.33	89.19	94.28	84.39	91.54
0.50	72.07	83.77	62.11	76.63	89.33	94.36	84.61	91.67
0.75	71.36	83.29	60.95	75.73	89.09	94.23	84.35	91.51
1.00	69.79	82.21	60.60	75.47	88.38	93.83	84.27	91.46

TABLE VI
QUANTITATIVE COMPARISONS OF THE PROPOSED METHOD WITH DIFFERENT BACKBONES ON THE SYSU-CD DATASET. THE BEST RESULTS ARE HIGHLIGHTED IN RED

Backbone	Params.(M)	FLOPs(G)	SYSU (%)		
			IoU	F1	Kappa
ResNet-18	17.59	24.92	72.07	83.77	78.93
ResNet-34	27.7	65.97	71.86	83.63	78.80
ResNet-50	31.47	73.04	70.29	82.55	77.55

6) Hyperparameter Sensitivity Analysis: We conducted a sensitivity analysis on the two key hyperparameters of learning rate and batch size. As summarized in Fig. 12, the configuration with learning rate of 0.0005 and batch size of 32 yields optimal and stable performance across all four datasets. Conversely, significant deviations from these values, particularly higher learning rates that cause training instability or smaller batch sizes that increase gradient variance, lead to a noticeable performance degradation. These findings collectively validate the effectiveness of our selected hyperparameter setup.

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V. Conclusion

This work identifies a critical limitation in existing CD methods, insufficient structural sensitivity for accurately identifying genuine changes via clear edge delineation. To address this, we propose S^2ENet for RS image CD. S^2ENet integrates the hierarchical adaptive FSCF, exploring high-frequency details in the frequency domain while performing contrast locking in the spatial domain, alongside the MBPF enabling deep cross-layer interaction and bidirectional feature completion. Extensive experiments on four benchmark datasets, comparing against nine SOTA methods through quantitative metrics and visual analysis, consistently demonstrate the superiority of the proposed S^2ENet framework.

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